Numerical Methods and Optimization

11/5/2026 – 3^{rd} Year

ENSIA — Spring 2026

Tutorial 11

- Metaheuristics -

Exercise 1: General Questions

(1) Give some exact methods of problem resolution.

(2) What is the objective of problem resolution?

(3) What is the difference between heuristics and métaheuristics?

(4) What is the difference between trajectory and evolutionary methods?

(5) How to study the efficiency of a metaheuristic

(6) What is Local Search? Tabu Search? Memoryless search?

(7) What is Aspiration criterion in tabu search method?

(8) Give the definition of Ant Colony Optimization ACO.

(9) True or False

- Simulated Annealing is more efficient than Tabu Search.
- With metaheuristics, we are guaranteed to find an exact solution for a given problem even if a local optimum is encountered
- Greedy algorithm finds the optimal solution

Exercise 2: Solving Traveling Salesman Problem with SA, TS and ACO

(1) Define the Traveling Salesman Problem and give its model

(2) Given an instance of 5 cities (in the figure below), What is the number of all possible solutions?

(3) Propose solution encoding for the TSP.

(4) We give the trivial initial solution: $x_0 = 1 - 2 - 3 - 4 - 1$, find all neighbors of x_0 using sub-tours inversion.

(5) Simulated Annealing

1. We want to implement simulated annealing with initial temperature $T = 20$, What is the probability of accepting each neighbor (found in the previous qst)?

2. Apply SA with two sets of parameters:
   $\Gamma_1 : T = 1000, T_f = 0.005, \alpha = 0.1$, and $\Gamma_2 : T = 100000, T_f = 0.005, \alpha = 0.05$.

3. Compare and study the results, Conclude.

(6) Tabu Search

1. Apply TS with two sets of parameters:
   $\Gamma_1 : LT = 10, K = 100$, and $\Gamma_2 : LT = 5, K = 100$.

2. Compare and study the results, Conclude.

(7) Ant Colony Optimization

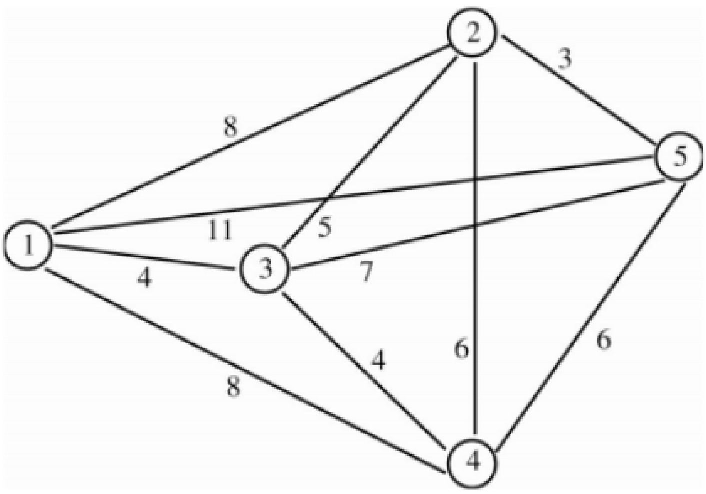
1. What is the equivalent of a solution in an Ant colony?

2. What is the equivalent of the communication between ants in ACO?

3. How many ants will be needed? What should each ant do and how?

4. Apply Ant Colony Optimization with the set of parameters
   $\Gamma : N = 50, K = 100, \tau_0 = 1, \rho = 0.95, \alpha = 0.5, \beta = 0.5$.

(8) Compare SA,TS and ACO, What do you conclude?



Exercise 3: Solving Assignment Problem using GA and VNS

(1) Define the Assignment Problem and give its model

(2) Given an instance of 5 persons and 5 tasks (C is the matrix of assignment costs), What is the number of all possible solutions?

$$C = \begin{pmatrix} 20 & 15 & 18 & 20 & 25 \\ 18 & 20 & 12 & 14 & 15 \\ 21 & 23 & 25 & 27 & 25 \\ 17 & 18 & 21 & 23 & 20 \\ 18 & 18 & 16 & 19 & 20 \end{pmatrix}$$

(3) Give a solution using the Greedy algorithm.

(4) Propose solution encoding for the AP.

(5) Ganetic Algorithm

1. What are the mutation and the crossover operators for this problem?
2. Apply Genetic Algorithm with the set of parameters $\Gamma : P_{size} = 50, p_m = 0.8, K = 100$.

(6) Variable Neighborhood Search VNS

1. Give $k_{max} = 3$ neighborhood structures to the shaking process
2. Propose a local search method.
3. Apply VNS with the set of parameters :$\Gamma : k_{max} = 3, K = 100$.

(7) Compare GA and VNS, What do you conclude?